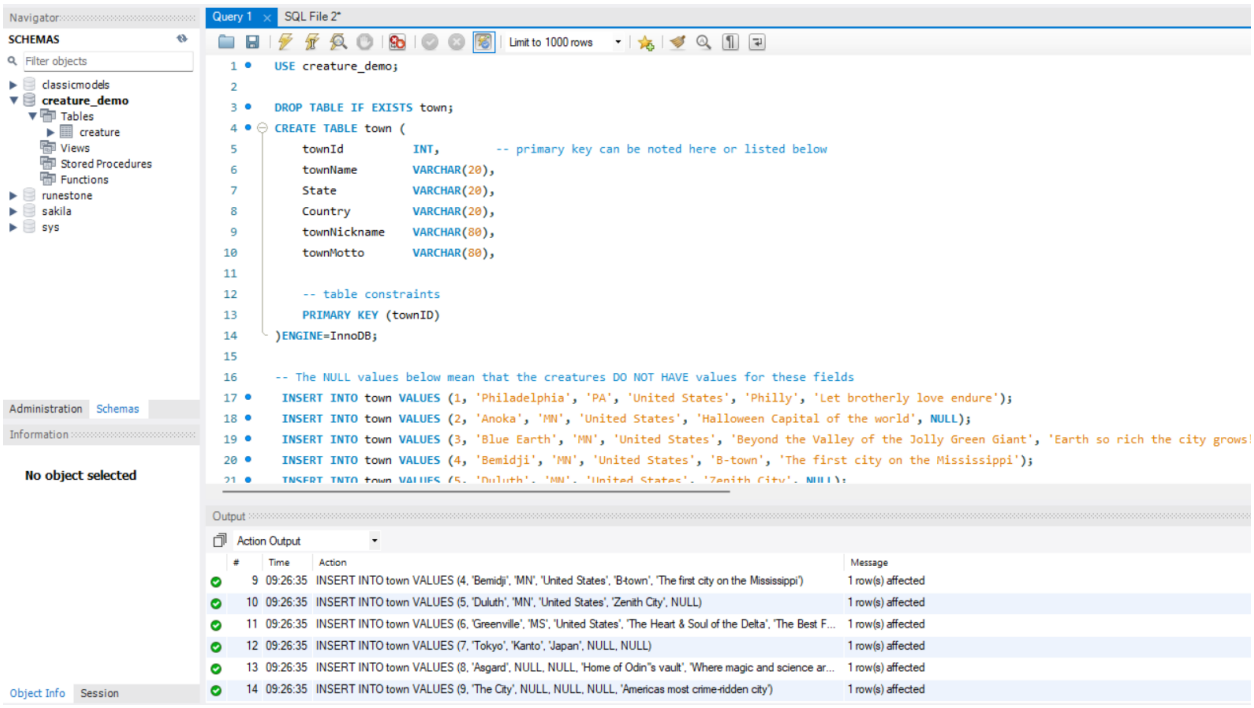
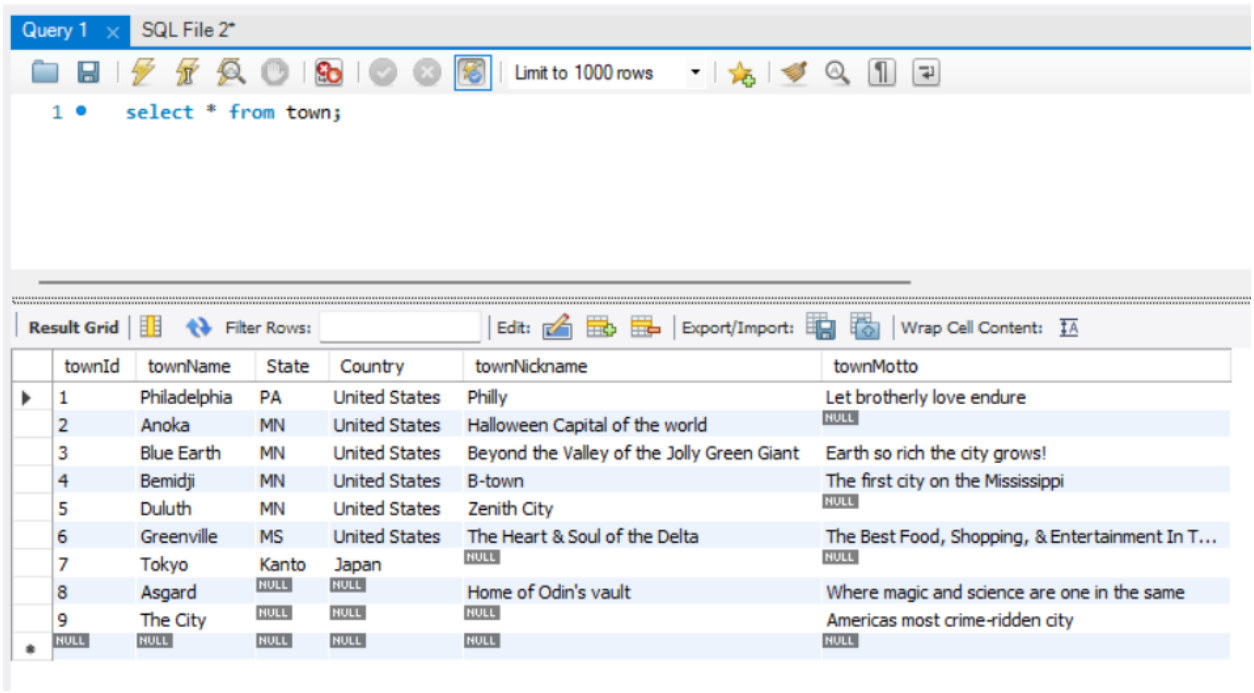
TASK ONE:

Created table ‘town’ in the creatures_demo database:

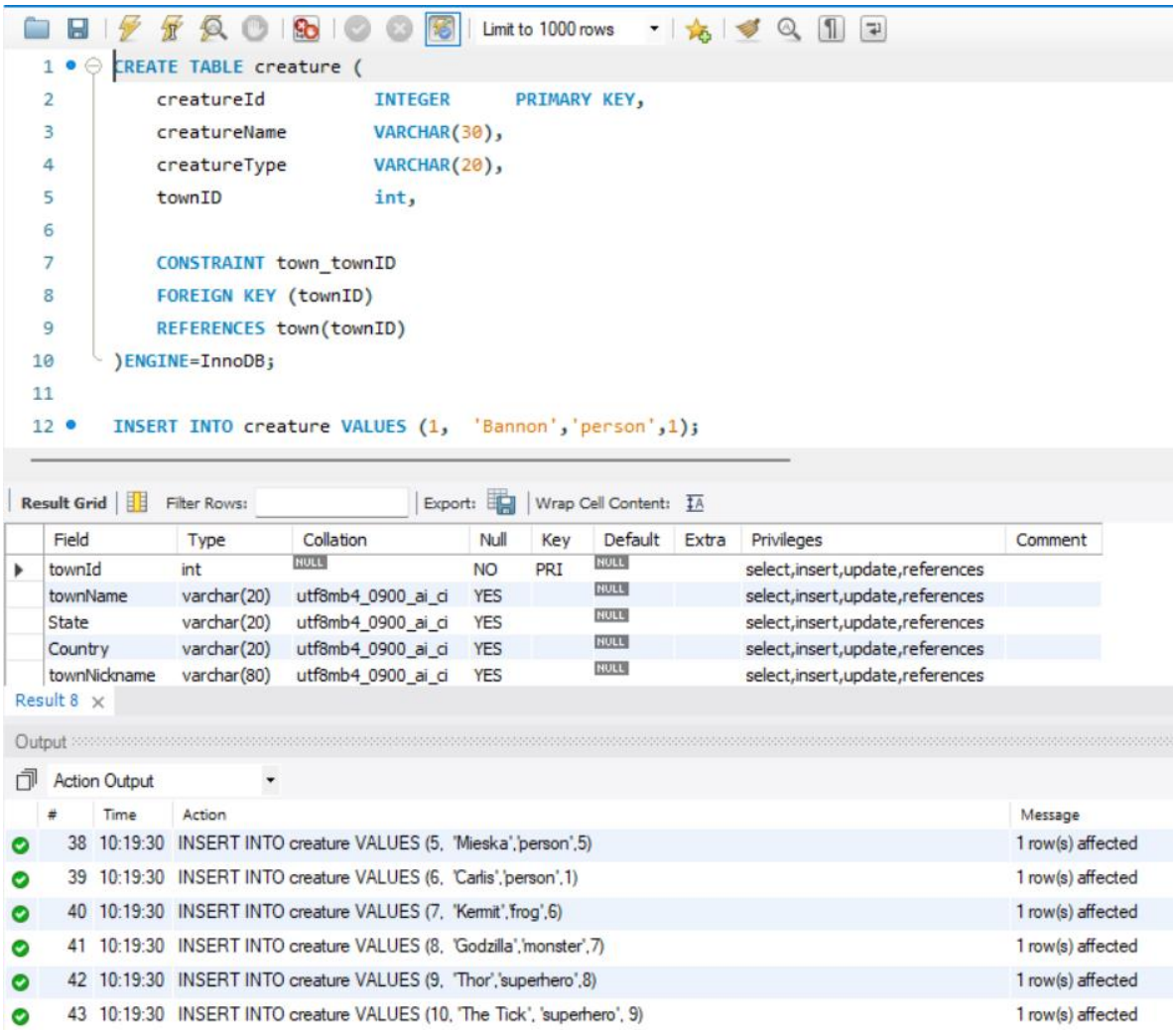




townId	townName	State	Country	townNickname	townMotto
1	Philadelphia	PA	United States	Philly	Let brotherly love endure
2	Anoka	MN	United States	Halloween Capital of the world	NULL
3	Blue Earth	MN	United States	Beyond the Valley of the Jolly Green Giant	Earth so rich the city grows!
4	Bemidji	MN	United States	B-town	The first city on the Mississippi
5	Duluth	MN	United States	Zenith City	NULL
6	Greenville	MS	United States	The Heart & Soul of the Delta	The Best Food, Shopping, & Entertainment In T...
7	Tokyo	Kanto	Japan	NULL	NULL
8	Asgard	NULL	NULL	Home of Odin's vault	Where magic and science are one in the same
9	The City	NULL	NULL	NULL	Americas most crime-ridden city
NULL	NULL	NULL	NULL	NULL	NULL

TASK TWO:

Recreate creature table:



```
CREATE TABLE creature (
    creatureId          INTEGER      PRIMARY KEY,
    creatureName        VARCHAR(30),
    creatureType        VARCHAR(20),
    townID              int,

    CONSTRAINT town_townID
    FOREIGN KEY (townID)
    REFERENCES town(townID)
)ENGINE=InnoDB;

INSERT INTO creature VALUES (1,  'Bannon','person',1);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Collation	Null	Key	Default	Extra	Privileges	Comment
townId	int	NULL	NO	PRI	NULL		select,insert,update,references	
townName	varchar(20)	utf8mb4_0900_ai_ci	YES		NULL		select,insert,update,references	
State	varchar(20)	utf8mb4_0900_ai_ci	YES		NULL		select,insert,update,references	
Country	varchar(20)	utf8mb4_0900_ai_ci	YES		NULL		select,insert,update,references	
townNickname	varchar(80)	utf8mb4_0900_ai_ci	YES		NULL		select,insert,update,references	

Result 8

Output

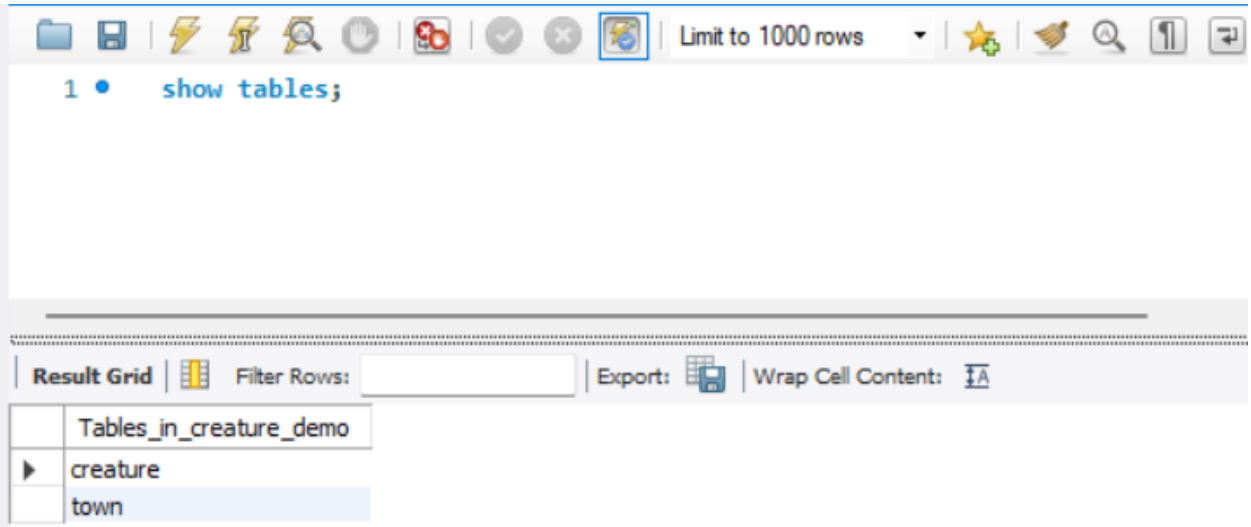
Action Output

#	Time	Action	Message
38	10:19:30	INSERT INTO creature VALUES (5, 'Mieska','person',5)	1 row(s) affected
39	10:19:30	INSERT INTO creature VALUES (6, 'Carlis','person',1)	1 row(s) affected
40	10:19:30	INSERT INTO creature VALUES (7, 'Kermit','frog',6)	1 row(s) affected
41	10:19:30	INSERT INTO creature VALUES (8, 'Godzilla','monster',7)	1 row(s) affected
42	10:19:30	INSERT INTO creature VALUES (9, 'Thor','superhero',8)	1 row(s) affected
43	10:19:30	INSERT INTO creature VALUES (10, 'The Tick', 'superhero', 9)	1 row(s) affected

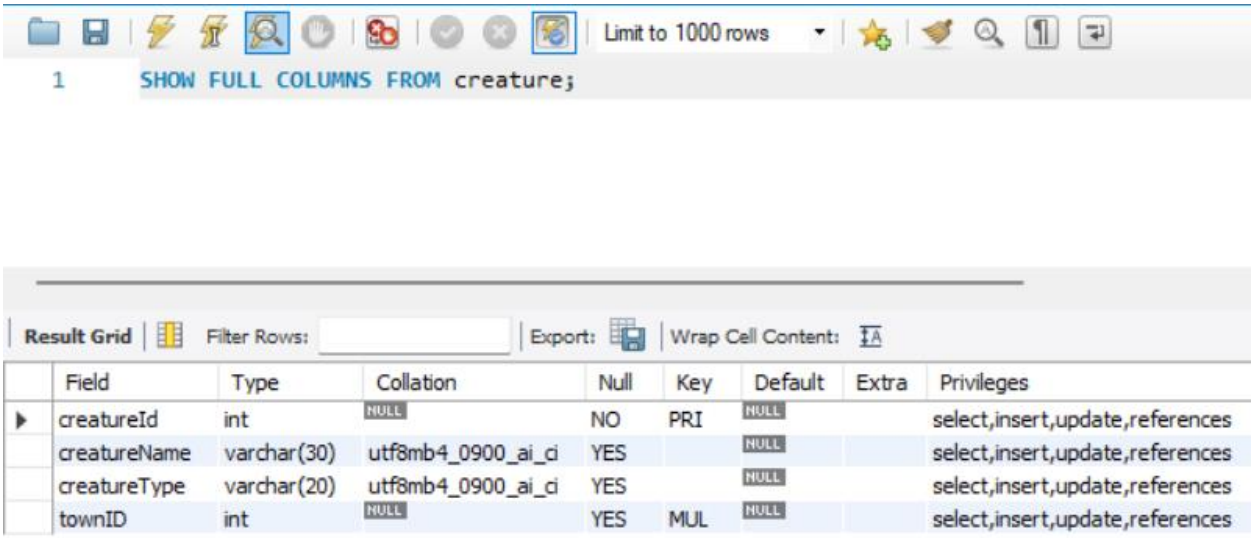
Can you explain why the Town table must be created and populated first? This may be a test question.

- The new create creature table references an attribute in the town table. This will produce an error if the town table doesn't exist in the database yet.

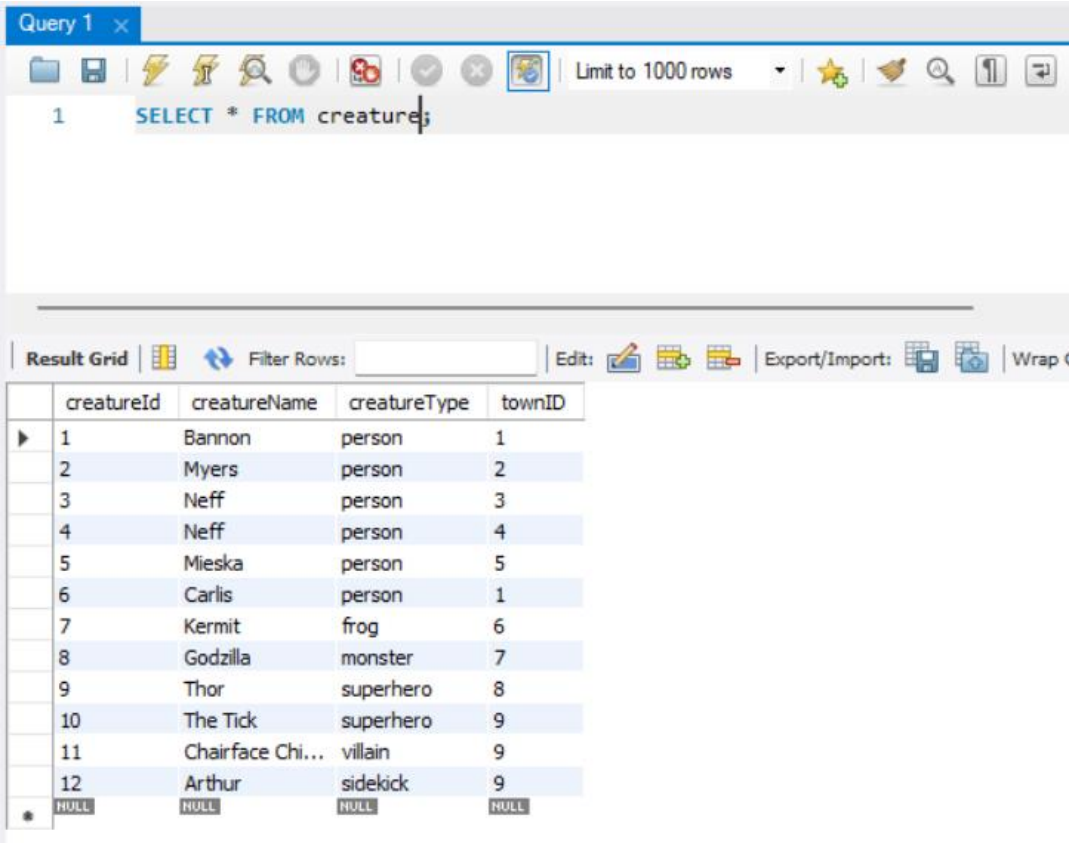
The creature database tables:



The structure of the Creature table:

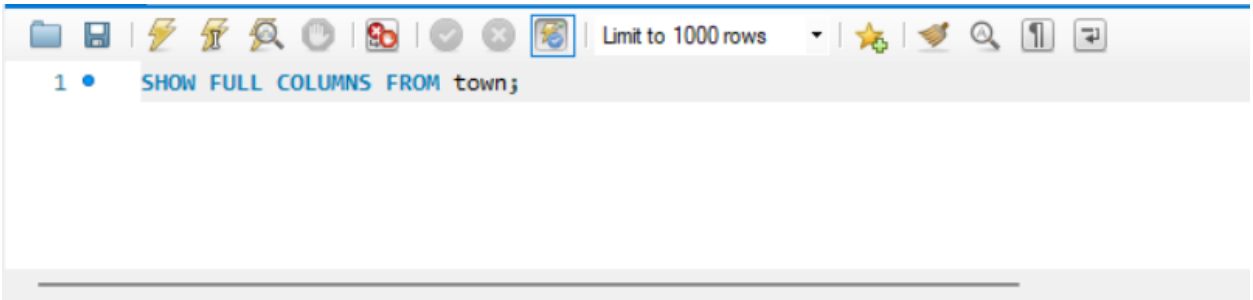


All of the Creature data:



creatureId	creatureName	creatureType	townID
1	Bannon	person	1
2	Myers	person	2
3	Neff	person	3
4	Neff	person	4
5	Mieska	person	5
6	Carlis	person	1
7	Kermit	frog	6
8	Godzilla	monster	7
9	Thor	superhero	8
10	The Tick	superhero	9
11	Chairface Chi...	villain	9
12	Arthur	sidekick	9
NULL	NULL	NULL	NULL

The structure of the Town table:



Field	Type	Collation	Null	Key	Default	Extra	Privileges
townId	int	NULL	NO	PRI	NULL		select,insert,update,references
townName	varchar(20)	utf8mb4_0900_ai_ci	YES		NULL		select,insert,update,references
State	varchar(20)	utf8mb4_0900_ai_ci	YES		NULL		select,insert,update,references
Country	varchar(20)	utf8mb4_0900_ai_ci	YES		NULL		select,insert,update,references
townNickname	varchar(80)	utf8mb4_0900_ai_ci	YES		NULL		select,insert,update,references
townMotto	varchar(80)	utf8mb4_0900_ai_ci	YES		NULL		select,insert,update,references

All of the Town data:

```
SELECT * FROM town;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

townId	townName	State	Country	townNickname	townMotto
1	Philadelphia	PA	United States	Philly	Let brotherly love endure
2	Anoka	MN	United States	Halloween Capital of the world	NULL
3	Blue Earth	MN	United States	Beyond the Valley of the Jolly Green Giant	Earth so rich the city grows!
4	Bemidji	MN	United States	B-town	The first city on the Mississippi
5	Duluth	MN	United States	Zenith City	NULL
6	Greenville	MS	United States	The Heart & Soul of the Delta	The Best Food, Shopping, & Entertainment In T..
7	Tokyo	Kanto	Japan	NULL	NULL
8	Asgard	NULL	NULL	Home of Odin's vault	Where magic and science are one in the same
9	The City	NULL	NULL	NULL	Americas most crime-ridden city
NULL	NULL	NULL	NULL	NULL	NULL

TASK THREE:

```
-- 1. Show all Creature names that live in 'United States'. Use United States as the search criteria.
select creatureName
from creature
join town on town.townID = creature.townID
where Country = 'United States';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

creatureName
Bannon
Carlis
Myers
Neff
Neff
Mieska
Kermit

```
-- 2. Show all creature names that do not live in the 'United States'
select creatureName
from creature
join town on town.townID = creature.townID
where Country <> 'United States';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

creatureName
Godzilla

```sql
-- 3. Show the motto for the town that The Tick resides in. Use The Tick as the search criteria
select townMotto
from town
join creature on town.townID = creature.townID
where creatureName = 'The Tick';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

townMotto
Americas most crime-ridden city

```sql
-- 4. Show all the "person" creatures that live in Minnesota (MN). Use 'person' and 'MN' as the search criteria.
select *
from creature
join town on town.townID = creature.townID
where creatureType = 'person'
and State = 'MN';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

creatureId	creatureName	creatureType	townID	townId	townName	State	Country	townNickname	townMotto
2	Myers	person	2	2	Anoka	MN	United States	Halloween Capital of the world	NULL
3	Neff	person	3	3	Blue Earth	MN	United States	Beyond the Valley of the Jolly Green Giant	Earth so rich the city grows!
4	Neff	person	4	4	Bemidji	MN	United States	B-town	The first city on the Mississippi
5	Mieska	person	5	5	Duluth	MN	United States	Zenith City	NULL

```sql
-- 5. How many creatures live in 'The City'? Use the COUNT function and 'The City' as the search criteria
select count(*)
from creature
join town on town.townID = creature.townID
where townName = 'The City';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

creatureId	creatureName	creatureType	townID	townId	townName	State	Country	townNickname	townMotto
2	Myers	person	2	2	Anoka	MN	United States	Halloween Capital of the world	NULL
3	Neff	person	3	3	Blue Earth	MN	United States	Beyond the Valley of the Jolly Green Giant	Earth so rich the city grows!
4	Neff	person	4	4	Bemidji	MN	United States	B-town	The first city on the Mississippi
5	Mieska	person	5	5	Duluth	MN	United States	Zenith City	NULL

```sql
-- 6. Show all the creature types that live in 'The City'. Use 'The City' as the search criteria
select distinct creatureType
from creature
join town on town.townID = creature.townID
where townName = 'The City';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

creatureType
superhero
villain
sidekick

TASK FOUR: SQL Insert statements and relationships

```sql
-- 1. Try to insert a new creature into the creature table.
insert into creature
values (13, "Bilbo Baggins", "hobbit", 10);
```

1. What happens when this query runs? Do you know why? Include your answer to this question in the submission document.



- It returns an error because the foreign key constraint will not allow an entry into townID that does not exist in the town table.

2. Insert a new town of your choosing. Include data for all fields.

```
1   -- 2. Insert a new town of your choosing. Include data for all fields.
2   insert into town
3   values (10, 'The Shire', 'ER', 'Middle Earth', 'The Shire', 'Peace and quiet and good tilled earth');
4
5   select * from town;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

townId	townName	State	Country	townNickname	townMotto
1	Philadelphia	PA	United States	Philly	Let brotherly love endure
2	Anoka	MN	United States	Halloween Capital of the world	NULL
3	Blue Earth	MN	United States	Beyond the Valley of the Jolly Green Giant	Earth so rich the city grows!
4	Bemidji	MN	United States	B-town	The first city on the Mississippi
5	Duluth	MN	United States	Zenith City	NULL
6	Greenville	MS	United States	The Heart & Soul of the Delta	The Best Food, Shopping, & Entertainment In T...
7	Tokyo	Kanto	Japan	NULL	NULL
8	Asgard	NULL	NULL	Home of Odin's vault	Where magic and science are one in the same
9	The City	NULL	NULL	NULL	Americas most crime-ridden city
10	The Shire	ER	Middle Earth	The Shire	Peace and quiet and good tilled earth
NULL	NULL	NULL	NULL	NULL	NULL

```
1   -- 3. Insert a new creature that resides in this town
2   insert into creature
3   values (13, "Bilbo Baggins", "hobbit", 10);
4
5   select * from creature;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

creatureId	creatureName	creatureType	townID
1	Bannon	person	1
2	Myers	person	2
3	Neff	person	3
4	Neff	person	4
5	Mieska	person	5
6	Carlis	person	1
7	Kermit	frog	6
8	Godzilla	monster	7
9	Thor	superhero	8
10	The Tick	superhero	9
11	Chairface Chi...	villain	9
12	Arthur	sidekick	9
13	Bilbo Baggins	hobbit	10
NULL	NULL	NULL	NULL

```sql
-- 4. Show the creature name, creature type, town name and town motto for the new creature
select creatureName, creatureType, townName, townMotto
from creature
join town on creature.townID = town.townID
where creatureID = 13;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

creatureName	creatureType	townName	townMotto
Bilbo Baggins	hobbit	The Shire	Peace and quiet and good tilled earth